

Statement of Work

BaseOps Usability Improvements for MRNMAP

BACKGROUND

The Volpe National Transportation Systems Center (Volpe Center) is the primary department of transportation agency that develops transportation related noise models needed to ensure the unobstructed utilization of the United States' transportation infrastructure. In this role, the Volpe Center provides technical support to other agencies, including the Federal Aviation Administration, the Federal Highway Administration and the Department of Defense (DoD). In particular, the Volpe Center is supporting the DoD in modeling military transportation related noise to meet AICUZ, NEPA and other mandates and statutes. In order to remove obstacles to general flight operations and flight training (needed for military readiness) around installations, the Volpe Center seeks to add new functionality to BaseOps to efficiently import and export data for military training exercises and to reduce conflicts between exclusion zones and air operations. BaseOps is a fundamental part of the DoD workflow for analyses. Since the required BaseOps software is owned by Wasmer Consulting, changes to the software must be made by them.

PURPOSE

The Volpe Center is seeking to engage the owner of the noise model BaseOps, Wasmer Consulting, to add functionality for the Department of Defense in support of DoD instruction 4165.57.

TASKING

Task 1. Add Spreadsheet-Like I/O Capabilities

The Contractor shall add functionality to import and export airspace operations in a manner similar to the existing functionality for flight and static operations. This functionality will relate to:

- Airspace Operations, Profiles and MOAs and MTRs
- MRNMAP noise model

Note: Altitude Bands are explicitly excluded from this functionality.

I/O Format for Airspace Profile Data

- Acceptable import format will be xml spreadsheet format consistent with existing BaseOps import functionality.
 - Excel spreadsheets can be saved in this format using the “save as” functionality.
- Acceptable export format will be xml spreadsheet format consistent with existing BaseOps import functionality.
 - Excel can read these files using the “open” functionality.

Selecting input/output (I/O) for Airspace Profile Data

The following features will be implemented to allow the user to select either import or export functionality for airspace operations:

- 1) Add import and export entries in the **File** menu (See Figure 1.)
 - a. **Import Airspace Operations from Spreadsheet ...**

- i. Opens the **Import Airspace Operations to Spreadsheet** Modal Window (See *Importing Airspace Profile Data.*)
- b. **Export Airspace Operations from Spreadsheet ...**
 - i. Opens the **Export Airspace Operations to Spreadsheet** Modal Window (See *Exporting Airspace Data.*)
 - ii. *Exporting Airspace Data.*

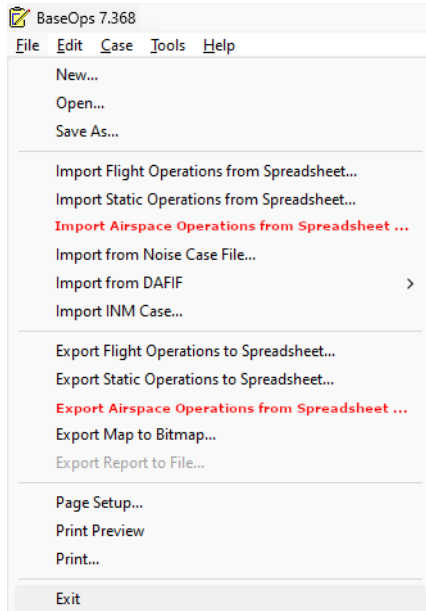


Figure 1: Added I/O Functionality for MOA Airspace Profiles – File Menu

Exporting Airspace Data

- 2) Create **Export Airspace Operations to Spreadsheet** Modal Window
 - a. Include **File** Tab (See Figure 2.)
 - i. Include default folder radio button
 1. Add **Export Airspace Operations to default spreadsheet file** radio button
 2. Default folder is determined in a manner consistent with other export options
 3. Displays default folder below radio button control
 - ii. Include select custom folder radio button with text control
 1. Add **Export Airspace Operations to the following spreadsheet file:** radio button
 2. Functionality consistent with other export options
 3. Add text control with browser functionality to select folder
 4. Displays selected folder in text control below radio button control
 - iii. Add text describing how to read and save to xml spreadsheet format.
 - b. Include **Columns** Tab (See Figure 3.)
 - i. Show list of mandatory parameters to export as columns in spreadsheet
 1. **Aircraft Name**
 - a. [Spreadsheet Column Name = Aircraft]
 2. **Airspace Profile Name**
 - a. [Spreadsheet Column Name = Profile]

3. **Area or Route Name**
 - a. The data in this column should be the name of the specific area or route in question.
 - b. [Spreadsheet Column Name = Area or Route]
4. **Annual Number of Day Operations**
 - a. [Spreadsheet Column Name = Annual Num Day]
5. **Annual Number of Night Operations**
 - a. [Spreadsheet Column Name = Annual Num Night]
6. **Annual Number of Evening Operations¹**
 - a. [Spreadsheet Column Name = Annual Num Eve]
- ii. Shows list of optional parameters with check box selection (See Figure 4.)
 1. **Airspace Profile Long Name**
 - a. [Spreadsheet Column Name = Long Name]
 2. **Airtime**
 - a. [Spreadsheet Column Name = Airtime, Sec/Op]
 3. **Airspeed**
 - a. [Spreadsheet Column Name = Airspeed, kts]
 4. **Standoff**
 - a. [Spreadsheet Column Name = Standoff, NM ref MOA Edge]
 5. **Engine**
 - a. [Spreadsheet Column Name = Engine]
 6. **Power**
 - a. *(Optional to include this pending LOE.)*
 - b. [Spreadsheet Column Name = Power]
 7. **Power Units**
 - a. (Wasmer to confirm it is OK to include this.)
 - b. [Spreadsheet Column Name = PU]
 8. **Noise Model**
 - a. [Spreadsheet Column Name = Noise Model]
 9. **Aircraft Category**
 - a. (Based, Transient, etc.)
 - b. [Spreadsheet Column Name = A/C Category]
 10. **Notes**
 - a. *(Optional to include this pending LOE.)*
 - b. [Spreadsheet Column Name = Notes]
- c. Include **Filtering** Tab (See Figure 4.)
 - i. Include **Export all Airspace Operations** radio button
 1. If selected, exports all airspace profiles when OK is clicked
 - ii. Include **Export only the Airspace Operations that match the following conditions:** radio button
 1. If selected, only exports airspace profiles meeting the conditions defined below in the text controls.

¹ If the BaseOps case uses evening data

- iii. Provide a list of text controls that allow selection of values for parameter defined in the label. Parameters include:
 1. Airspace Profile Names
 2. Airspace Profile Long Names
 3. Area or Route Names
 4. Aircraft Names
 5. Engine Names
 6. A/C Categories
 7. Added Groups
 - a. (Any groups added under the baseops “Groups” section that are applied to airspace profiles, e.g. Noise Models, Sources, etc.)
- iv. Clicking on the three dots at the end of each text control opens filter dialog modal window that allows for selection of conditions to match (See Figure 5.)
 1. Select All, Select None and Invert Selection behave as names suggest
 2. Once OK is clicked the selected conditions replace any existing conditions for the specified parameter.

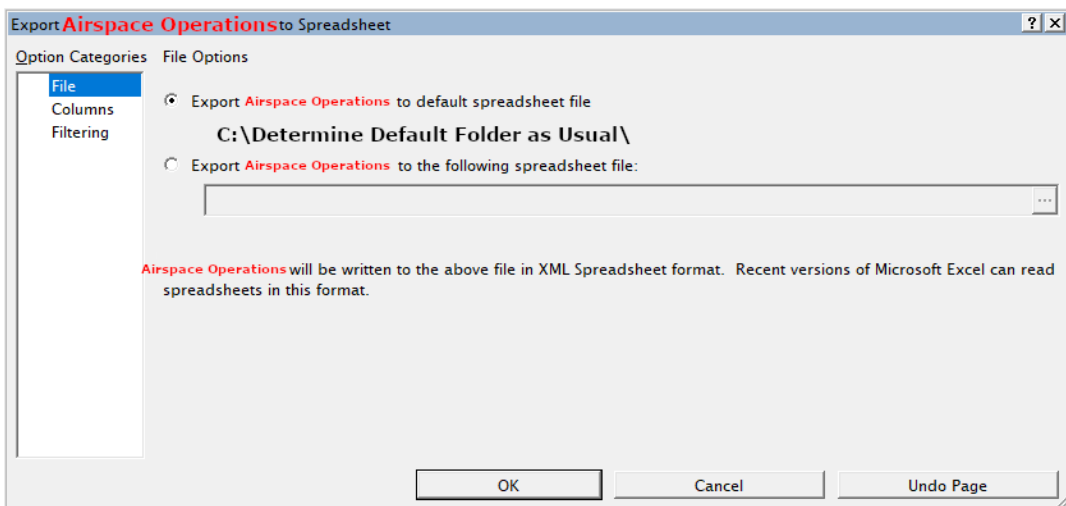


Figure 2: Added I/O Functionality for Airspace Operations – Export Options - File

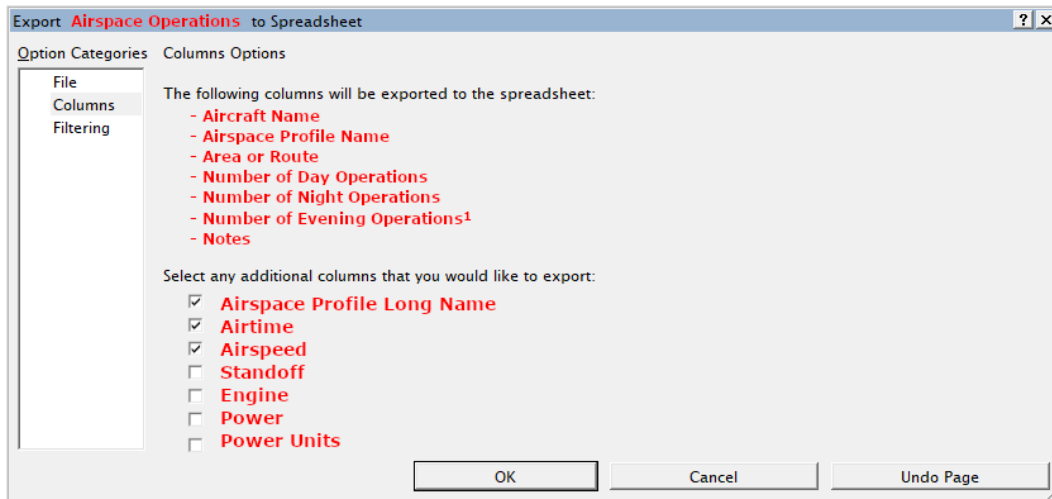


Figure 3: Added I/O Functionality for Airspace Operations – Export Options - Columns

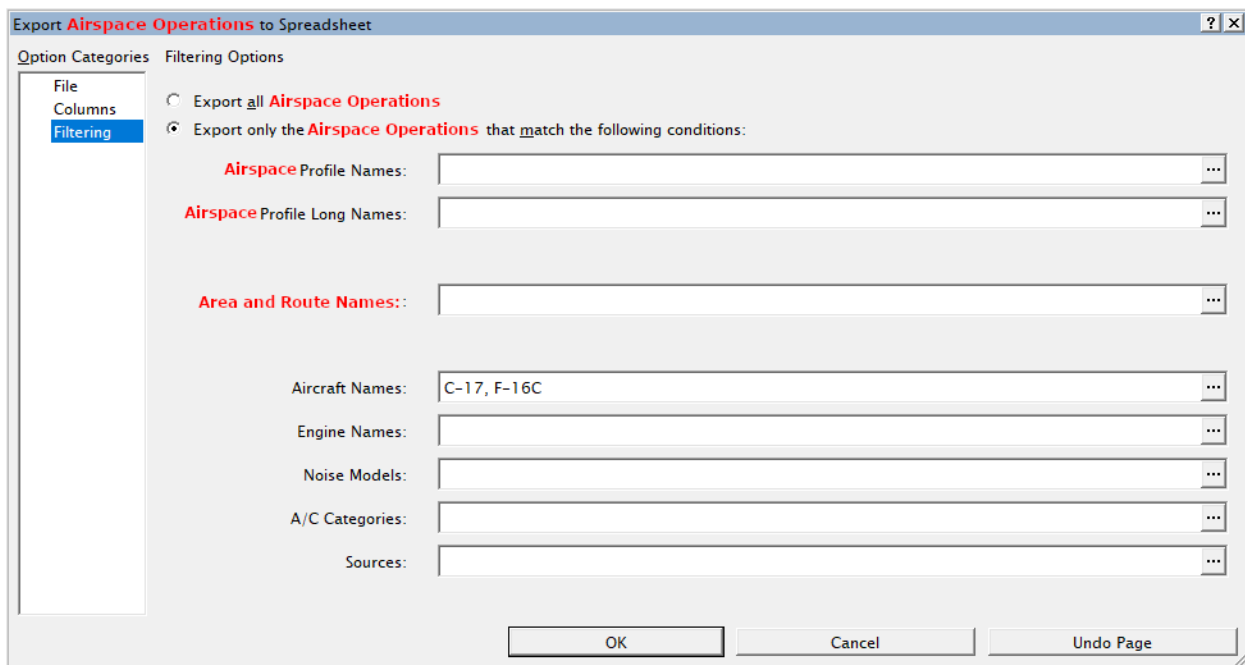


Figure 4: Added I/O Functionality for Airspace Operations – Export Options - Filtering

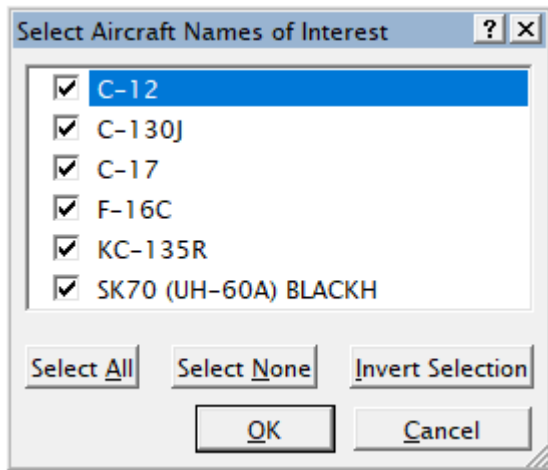


Figure 5: Example of Export Filter Condition Selection in Filter Dialog

Importing Airspace Profile Data

3) Create **Import Airspace Operations from Spreadsheet** Modal Window

- a. Include **File** Tab (See Figure 6.)
 - i. Include default folder radio button
 1. Add **Import Airspace Operations from default spreadsheet file** radio button
 2. Default folder is determined in a manner consistent with other export options
 3. Displays default folder below radio button control
 - ii. Include select custom folder radio button with text control
 1. Add **Import Airspace Operations from the following spreadsheet file:** radio button
 2. Functionality consistent with other export options
 3. Add text control with browser functionality to select folder
 4. Displays selected folder in text control below radio button control
 - iii. Add text describing how to read and save to xml spreadsheet format.
- b. Include **Columns** Tab (See Figure 7.)
 - i. Show list of automatic parameters to import as columns in spreadsheet. Include drop down menu to the right showing the default column but allowing selection of a different column.
 1. **Airspace Profiles Name Column**
 - a. [Spreadsheet Column Name = B]
 2. **Annual Number of Day Operations**
 - a. [Spreadsheet Column Name = D]
 3. **Annual Number of Night Operations**
 - a. [Spreadsheet Column Name = E]
 4. **Annual Number of Evening Operations²**

² If the BaseOps case uses evening data

- a. [Spreadsheet Column Name = F]
 - ii. Shows list of optional parameters with check box selection (See Figure 7.)
 - 1. **Area or Route Name**
 - a. [Spreadsheet Column Name = C]
 - 2. **Notes**
 - a. [Spreadsheet Column Name = G]
- c. Include **Missing Data** Tab (See Figure 8.)
 - i. Handle missing airspace profile in BaseOps file when spreadsheet has the data.
 - ii. Handle missing airspace profile in spreadsheet when BaseOps has the data.
 - iii. Handle missing Area or Route in BaseOps file when spreadsheet has the file AND Areas or Routes are being imported.

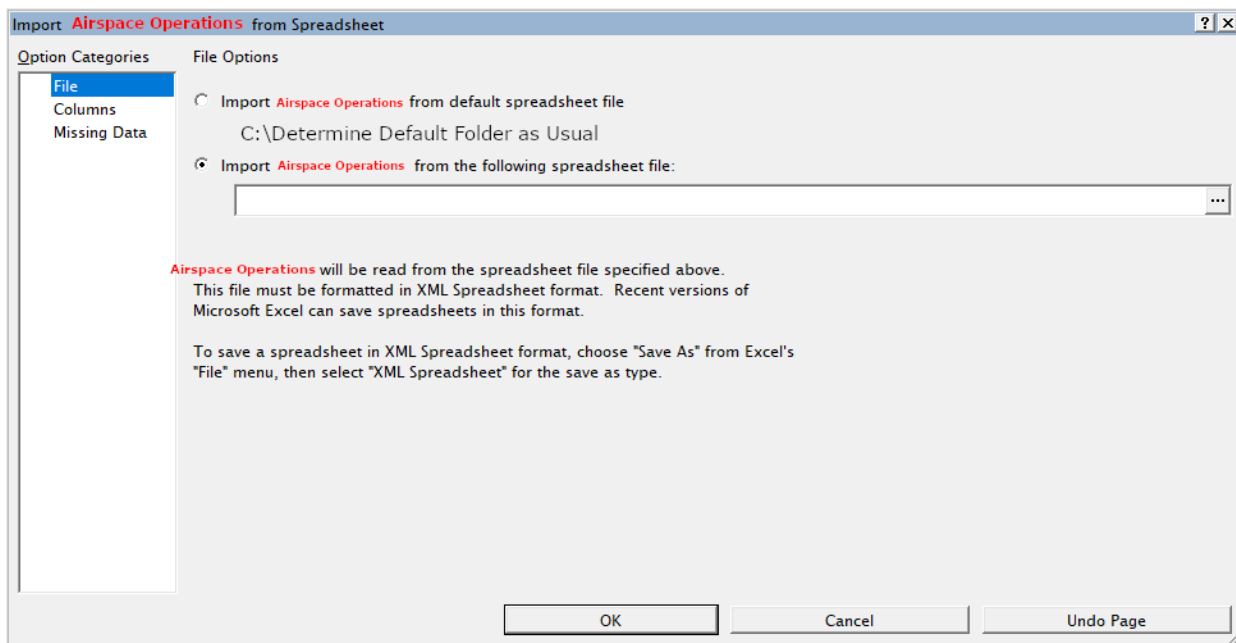


Figure 6: Added I/O Functionality for Airspace Operations – Import Options - File

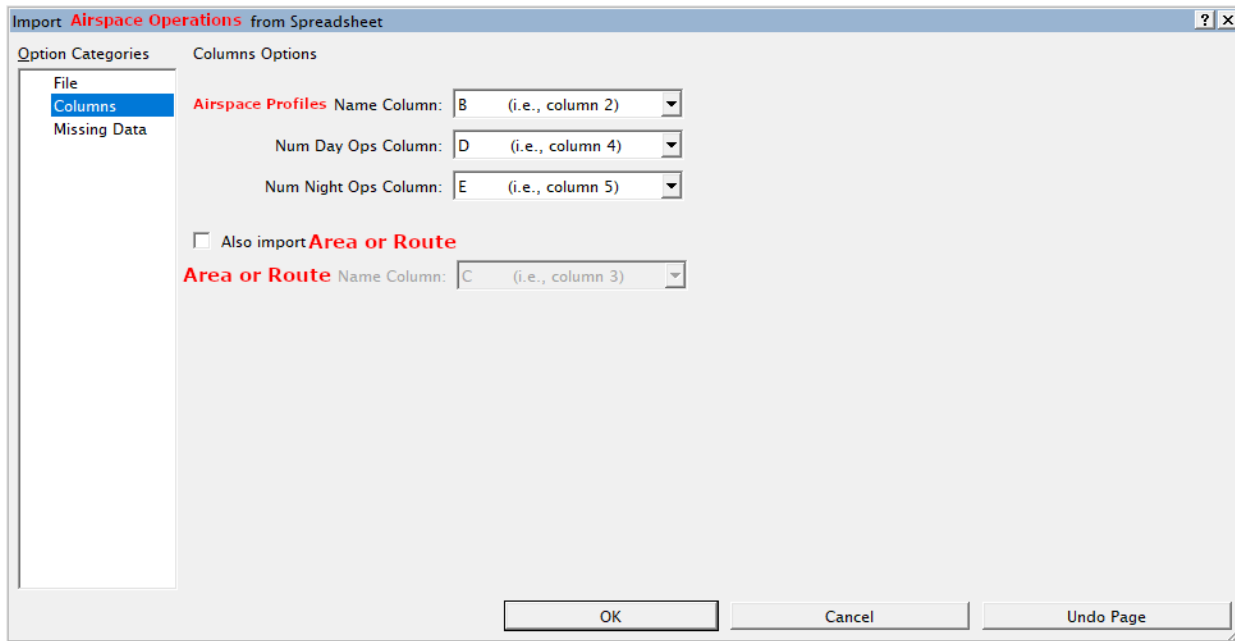


Figure 7: Added I/O Functionality for Airspace Operations – Import Options - Columns

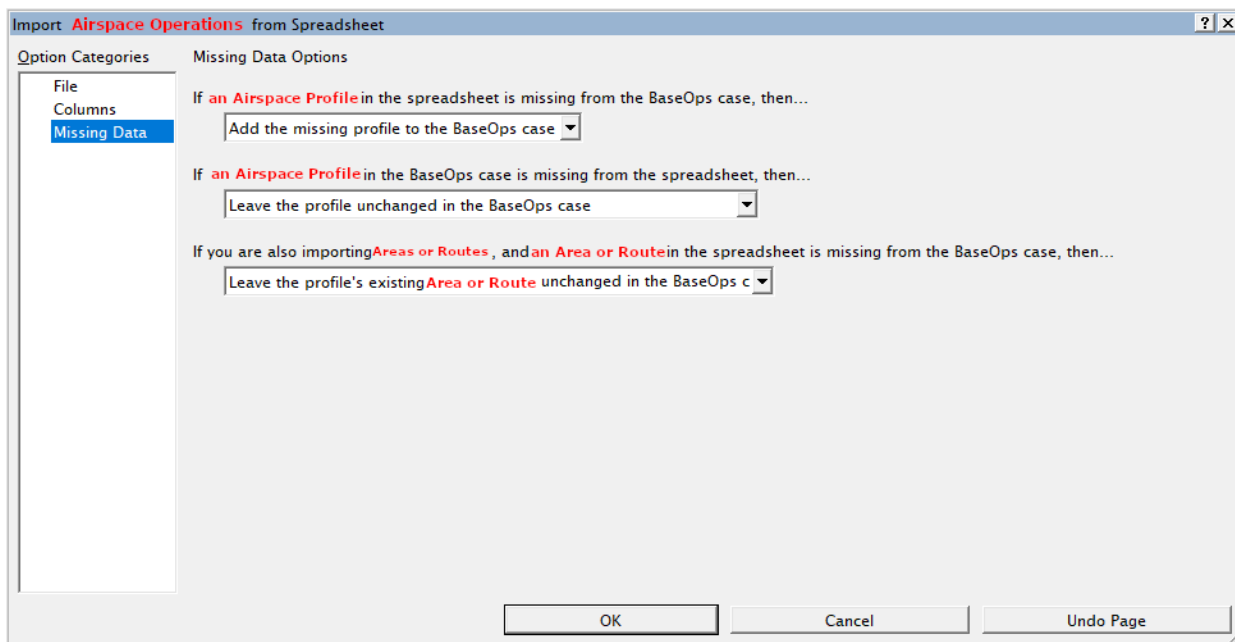


Figure 8: Added I/O Functionality for Airspace Operations – Import Options – Missing Data

Task 2. Improve “Track / Avoidance Area” when MOAs/MTRs intersect Avoidance Areas (AAs) for MRNMAP

- Add a user warning when using the Check Case functionality that would warn the user when an Avoidance Area/Military Training Route intersection existed.

<u>CLIN</u>	<u>Task(s)</u>	<u>Deliverable</u>	<u>Due Date</u>
00100	1	Added functionality to import and export airspace operations replicating the existing functionality for flight and static operations.	03/12/2026
	1	User Documentation	03/12/2026
00200	2	Added track / avoidance warning functionality	03/12/2026
	2	User Documentation	03/12/2026

ACCEPTANCE CRITERIA

To be accepted, deliverables shall be emailed/ a link provided to the project technical POC, Aaron Hastings, at Aaron.Hastings@dot.gov. The deliverables will be reviewed for compliance with the SOW, timeliness, and format. The POC will acknowledge acceptance of each deliverable by email.